CHANNEL PROTEINS DRIVE MICELLAR INTERACTIONS WITH THE BACTERIAL OUTER MEMBRANE

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ABSTRACT:

Recent studies report the abundance of integral membrane proteins in the outer membrane of Gram-negative bacteria [1], challenging the conventional lipid-centric models of membrane organization. This revised understanding of the architecture of the bacterial membrane necessitates a re-examination of how antimicrobial molecules interact with them and eventually permeate this complex biological barrier. Surfactants are one of the most widely used class of antimicrobials in daily life, whose interactions with membranes are complicated by their tendency to aggregate as micelles [2,3]. Using multiscale molecular dynamics simulations, we observe a preferential affinity for micelles to bind to channel proteins in the outer membrane, over lipids. Micelles are observed to bind to the extracellular loops of the channel protein OmpF to result in obstructed channelling of other small molecules. Interestingly, surfactants eventually break free from micelles in this bound state to move further into the channel, suggesting a mechanism through which surfactants permeate through porins despite initially assembling into micelles substantially larger than the pore diameter. We further examined the energetics of complete translocation of a surfactant through the channel by estimating the free energy associated with the process. Using the finite temperature string method, the free energy barrier observed for crossing the constriction region of the channel is examined in a multidimensional collective variable space. These calculations confirm significant affinity for surfactants to enter the channel, while crossing of the constriction region requires conformational acrobatics in the form of folding and flipping leading to a barrier crossing event. Using the above approach, our study provides new insights on how aggregation and structural properties of surfactants lead to differences in mechanism for them to translocate complex membrane barriers.

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